You said:

. It installs!
2. Settings work! Love the cogwheel and file solition.
3. Importing map works but does not change to viewport shading. Second map
imports and sucessfully hides the first.
4. switching between maps works great
5. Paint texture works but klicking it again does not exit paint mode and while
in paint mode paint lightmap is greyed out. Pressing tab works of course to get
back to edit. Same with paint lightmap. acessable from edit mode but the
buttons dont switch between paint styles or exit paint mode.
switching painted texture off reveals the same texture without paint but also
with weirdly wrong uv. Wall textures seem translated up or downwards and floor
textures are zoomed. That mus be fixed. Also another button for disabling
textures to see the pure lightmaps would be good.
could not switch painted lightmaps: name
*_restore_non_quake_lights
is not defined.

6. Exporting automatic an both works

The lithing panel does not fully work but i guess thats for the next round.

Thinking

ChatGPT said:

Excellent! That's a very useful first GUI test. The important foundations are working: installation, configuration, map collections, map switching, and both export modes. We also have several concrete issues rather than vague complaints.

I'd make the next release a focused **v0.18.1 painting and viewport-correction update**, rather than expanding the feature set.

I've identified five things to address:

1. Switch to the correct viewport shading automatically after import.
2. Make paint buttons behave as actual toggles and allow direct switching between paint modes.
3. Correct the original texture UVs when painted overrides are disabled.
4. Add a texture visibility switch so lightmaps can be inspected without diffuse textures.
5. Fix the undefined `_restore_non_quake_lights` reference breaking painted-lightmap switching.

And we'll preserve the working multi-map and export behaviour.

You said:

you still there?

Worked for 5m 46s

ChatGPT said:

Good test report! I've turned those findings into **QSS v0.18.1**. The fixes are committed to our existing feature branch, and the CI build has passed.

QSS v0.18.1 CI passed

What changed

1. **Paint buttons now behave as toggles.** Clicking the active button exits paint mode and returns to the previous Object or Edit Mode. Clicking the other button switches directly between texture and lightmap painting.
2. **Paint controls remain accessible during Texture Paint.** Corrected Blender's context-mode detection, which caused the disabled button problem.
3. **Original texture UV restoration is repaired.** When painted textures are disabled, QSS reconstructs the original Quake texinfo projection from the source BSP instead of reusing the face-local painting UVs. This specifically addresses the shifted wall textures and zoomed floor textures.
4. **New Diffuse Textures switch.** Turn it off to inspect the lightmaps without the base texture contribution. It works across the lighting presets.
5. **Painted-lightmap switching no longer calls the missing helper.** Removed the stale reference that caused your reported exception.

6. **Viewport shading updates target the active 3D View space** and trigger a redraw.

Install v0.18.1

Blender-ready ZIP Build #384

Download QSS v0.18.1

SHA-256: `f1a3b82de56968cf9c76767c4857f9a0fd9f63023d77cf4be1531a44a90f1886`

View successful CI run #384 ↗

CI passed compilation, standalone tests, version verification, package assembly and artifact upload.

What I'd like you to test next

I'd prioritise these four checks:

- Enter Texture Paint, click Paint Texture again, and confirm it returns to Edit Mode.
- While painting, click Paint Lightmap and confirm it switches targets without requiring Tab.
- Disable Painted Textures on a map where you've already painted some faces. Check the wall and floor UV alignment.
- Switch Painted Lightmaps on and off in Original and Baked modes.

Also check whether importing now switches the viewport to Material Preview.

I haven't claimed interactive Blender verification yet. The next GUI test will tell us whether these corrections behave as intended.

PR #1 is updated and remains draft and unmerged. `main` is untouched.

We're getting closer to that pleasant, predictable editing workflow. 😁

You said:

when importing a map also set the edit mode to select faces. just more
convenient.

the paint buttons work as intended now. good job.

I guess we need the isolate faces feature again. Its hard to paint in
tight spaces. :D Just put it up in the navigation section. And rename the
navigation secton to "Quick Actions" The four buttons here ([WASD},[Move to
playerstart],[isolate selection], [focus on face]) should be all horizontally in
a row. and if possible i want each to have a different background color.
Isolating should ideally work in edit and in paint mode. Anoter button would be
[focus on face] it puts the camera on the axis of the faces normal on a distance
so it fits neatly in the viewort and temporarily adusting the renderdistances so
nothing between the camera and the face is drawn. The renderdistances gets
switched back the moment the paint mode is left, another face is focussed or the
camera is moved.

disabling painted textures works great... kind of. I only notice it when
switching it off and on but the painted texture seems to be a tiny bit warped.
As if they had slightly different mapped uv. I meen the uv is in the right spot
now but dunno reminds me a bit like two images downscaled with different
algorithms. Not a big deal but real. :D Also we have already the three buttons
to toggle diffuse tex painted tex and painted lightmap. Put the diffuse
lightmap there too. Put the buttons in a table. Rows are Textures and Lightmaps
and Colums Painted and Diffuse (is source better? What will be intuitively
undestood). And let the user check and uncheck the boxes.

switched to cycles with the button. Works. Clicked bake:

Cycles bake failed; atlas and light calibration restored: Cycles finished but
sampled BSP lightmap luxels are all zero. Energy scale=1; imported lamps=149;
max lamp energy=300 W; sampled luxels=218901; whole atlas nonzero
pixels=3/4194304; whole atlas max RGB=0.000830644. Use Save Lighting Diagnostics
and send the report; failed-bake statistics are recorded.

Worked for 11m 27s

ChatGPT said:

Good news: **QSS v0.18.2 is built, and CI passed.** We've got the next iteration ready for Blender 5.2 testing. :D

QSS v0.18.2 Build passed

What's changed

Import

- A successfully imported BSP now opens in Edit Mode with Face Select active.
- The player-start view is applied across open 3D Views.

Quick Actions

- Replaces NAVIGATION with one horizontal row: WASD, Start, Isolate and Focus.
- Face isolation works in Edit Mode and Texture Paint, preserving the previous hidden faces and selection.
- Focus aims along the selected face normal and frames it closely.
- During painting, temporary clipping removes foreground occluders. Original clipping distances are restored when you leave Paint Mode, move the view or focus another face.

Lighting display matrix

The three previous switches are now a two-row matrix:

	Painted	Source

Textures	Painted texture	Original diffuse texture
Lightmaps	Painted lightmap	Unpainted active lightmap

I chose **Source** rather than Diffuse because it describes both rows consistently.

Texture precision

- Padded Quake-valid texture dimensions no longer stretch the face's UV footprint.
- Original texel density is preserved for face-local images up to the 512-pixel limit.
- Larger images are resampled only when necessary.
- Added regression tests for both cases.

Cycles bake

- Bake visibility is now restricted to the active map's lights.
- The mesh and its lights are temporarily made render-visible, with their previous visibility restored afterwards.
- Failed-bake diagnostics contain additional visibility information.

The bake change is a targeted investigation, not yet a confirmed fix. Your previous zero-luxel result needs another Blender test.

Download

Blender install ZIP CI #410 passed

Download QSS v0.18.2

View CI run #410 ↗

SHA-256: `c92a6ef1c3dccb7c8debad9e0c3822695b4fa9128b5b8af008c0876e1aeacb7e`

One limitation: Blender's standard Python sidebar API doesn't provide arbitrary per-button background colors. I've used distinct icons and compact labels instead.

The PR is updated and remains draft/unmerged. `main` is untouched. CI does not test interactive Blender behavior.

The next GUI test

I'd especially like to know whether:

- Import reliably lands in Face Select Edit Mode.
- Isolate works without disrupting Texture Paint.
- Focus and clipping behave naturally in tight spaces.
- The texture mapping now looks identical when switching Painted on and off.
- The new matrix is comfortable to use.
- The Cycles bake produces nonzero luxels.

Should the bake still fail, please use **Save Lighting Diagnostics** and send me the report. That will give us substantially more information about where the light transport is being lost.

We're getting a proper surface-editing workflow now. 😁

You said:

Love that it goes to edit mode and choses face selection when importing now.
Should also deselect all in the same moment tho to avoid a completely orange
screen because everything is selected. :D

Is it normal, that in edit mode everything is flat lit? Black gets real black
when changing to object. mode. Would prefer it with the normal color space in
edit mode too.

Definitely starts to look like a usable tool! Good job
selected a face, clicked paint texture:
Python: Traceback (most recent call last):
  File
"/home/chnucki/.var/app/org.blender.Blender/config/blender/5.2/scripts/addons/
quake_surface_studio/addon.py", line 5409, in execute
    result = bpy.ops.qss.detach_selected_faces()
RuntimeError: Error: Could not detach face (prepare face-local paint images): A
face-local projection requires 8 axis values, 4 bounds and 2 output dimensions..

If the mesh was hidden/revealed in Edit Mode, switch to Object Mode and
re-import if the face/UV attribute counts are stale.

paint lightmap works tho

Isolate face works fantastic.

focus on face works well exept for the snug fitting. :D The camera should move
far enogh away to get the whole "bounding box" of the face on screen. Also it
would be nice if the camera translates to the new position in a lets say .5 sec
travel to make it less desorienting. Also two things: If pressed again while
focussed on a face it moves to the next selected one. And if really cool
hovering over the button and mouseweeling could flip thrugh selected faces back
and forth.

Worked for 7m >

ChatGPT said:

Great to hear that **Face Isolate** and the import-to-Face-Select workflow are behaving well.

I found a concrete cause for the Paint Texture crash: the face-local projection helper expects eight numeric texinfo values, but the detach code passes the parsed BSP texinfo record directly. The failure happens while preparing the painted face image, before Texture Paint starts; it is separate from the Edit Mode selection issue. I'm correcting that input contract rather than weakening the validation.

For the import, I'll also clear the BMesh selection after entering Face Select, so the map opens without the orange selection overlay.

CI passed **QSS v0.18.3 is ready for Blender 5.2 testing.**

Download QSS v0.18.3 — Blender install ZIP ↗

SHA-256: `f99f05075224007a7d43d6292f3442c6ae549fa85017e7c94d98687d7f3a21ee`

Changes made

- **Import:** enters Face Select Edit Mode and clears all face, edge and vertex selection.
- **Paint Texture:** fixed the texinfo input mismatch. BSP29 stores eight projection floats followed by two metadata integers; the helper now accepts the full record. Added a regression test for it.
- **Focus:** fits the projected face bounds to the viewport with more breathing room.
- **Camera movement:** eases location, rotation and distance over 0.5 seconds.
- **Face cycling:** clicking Focus again advances through selected faces. The mouse wheel over the Focus button cycles forward/backward.

The existing clipping restoration behavior is retained. The wheel listener and animated camera are new interactive behavior, so they still need a real Blender test.

About the Edit Mode lighting

That is not supposed to be a different color space. Blender's **Solid** → **Lighting** → **Flat** mode deliberately calculates no lighting, whereas **Material Preview** uses Eevee and an HDRI and is intended for material preview and texture painting. These shading modes apply across viewport modes. Blender 5.2 LTS...

QSS already requests Material Preview for the Original/Baked view and does not change the scene's view transform. So I haven't claimed this is fixed in v0.18.3: first check `Z` → `Material Preview` while in Edit Mode. If it is already selected and the appearance still changes when switching modes, a screenshot of the viewport shading popover in Edit Mode would help isolate the cause.

Validation

GitHub Actions run #422 passed compilation, standalone tests, version checking, package assembly and artifact upload. It does not exercise Blender's interactive viewport, so please specifically test Paint Texture, import deselection, Focus framing/animation, and wheel cycling.

PR #1 — v0.18.3 ↗ is updated and remains draft. `main` is unchanged.

You said:



Focus beautifully moves but seeing it id say it should stop a bit farther away
to leave lets say a 10% margin for foreground ui stuff. Also sometimes it
approaches the normal from the wrong side looking at the face from behind. :D
That has to be fixed but besides that: use that camera behaviour also for move
to player start and any possible future functions moving the camea somewhere.

why must i be in object mode to use the rexture lightmap matrix?

Painted textues dont warp anymore! Good job! The table with the checkboxes
works but is pretty off layout wise. Its obvious in the picture. Any ideas?

The wasd movement is great. any idea for an unused shortcut that is easily
accessable with the left hand over wasd?

I would like to be able to also use an external editor to edit textures and
Lightmaps. Thinking of krita.

Selecting too many faces and entering paint mode can crash blender. There
should be a warning threshold when someone trys to start paint mode with more
then 64 faces selectred.

I attached picture. Object mode left, Edit mode right.

Some questions:

Quake has a 256 color palette. Are we converting to it at some point? I ask becuase it seems i can paint with any color.

Exporting a map doesnt take very long since all the assembly is already done
when export is used am i right? Cause if that is the case we can just allow
[test in ironwail] all the time and just export a short temporary map that is
deleted afterwards.

Worked for 5m 34s

Analyzed

ChatGPT said:

QSS v0.18.4 — CI passed

Download QSS v0.18.4 for Blender 5.2 ↗

SHA-256: `d78852c01ff4ec240afb33ac500a5e1cb3fd697f287b388989964f45695ab112`

Draft PR #1 ↗ is updated. It remains unmerged; `main` is unchanged.

What changed

Focus on Face: the camera now chooses the side of the face occupied by the current view, instead of blindly following a possibly reversed normal. I also increased the framing margin to give the face more breathing room.

Camera travel: Player Start now uses the same 0.5-second eased camera transition. That transition is a reusable QSS operator for future camera-moving actions.

Layer matrix: fixed-width columns should align the row labels and checkboxes more cleanly. The matrix is now available in Edit Mode as well as Object Mode. Painted Textures briefly switches to Object Mode internally for its mesh/UV changes, then restores Edit Mode.

Paint safety: both paint entry points now reject selections above 64 faces with a warning. Split larger selections into batches.

Ironwail: there is now a separate **Test in Ironwail** action. It needs a configured executable and game directory, but no previous export. It exports a temporary copy without overwriting your normal export, then removes the temporary BSP and optional `.lit` when the launched Ironwail process exits while Blender is running. Closing Blender first can leave the temporary file in `maps/qss_preview`.

CI passed compilation, standalone tests, version validation, package assembly and artifact upload. The actual camera behavior, Edit Mode toggles, paint guard and Ironwail launch still need your interactive Blender test.

Your questions

Quake's 256-color palette

Yes, but **not while you're painting**.

QSS imports palette indices as ordinary RGB Blender images. You can paint with any RGB color. When exporting a BSP, QSS converts the pixels to the closest available Quake palette entry and writes indexed miptexture data. Quake's texture format uses palette indices into a 256-color table. qamers.org

That is intentional: it makes painting pleasant and preserves the Quake format at the export boundary. Lightmaps are a separate path; they are not restricted to the texture palette in the same way.

External editing with Krita

This is a good next feature. I'd implement a proper round-trip rather than merely opening an image externally:

- **Textures:** export the active texture as PNG, open in Krita, then reload the saved pixels into the *same Blender image datablock*. That preserves materials and UV references.
- **Lightmaps:** use 32-bit OpenEXR to preserve floating-point lighting values rather than clipping them into an ordinary 8-bit PNG. Krita supports OpenEXR paint-over and float interchange. docs.krita.org +1
- Keep the original image dimensions fixed during the round-trip initially. Resizing an atlas would invalidate UV placement.
- Add explicit **Open Externally** and **Reload from Disk** actions, with the external file path tracked per image.

One important color-management detail: EXR doesn't embed its color space, so Blender and Krita need matching OCIO/profile settings to avoid visual shifts. docs.krita.org

A shortcut for WASD navigation

My candidate is **Shift+W**. It's easy to reach without moving your left hand away from WASD. I wouldn't claim it is universally unused, because custom keymaps and add-ons can add bindings. Blender's own default Walk/Fly shortcut is `Shift` + the accent-grave key.

Blender 5.2 ... +1

Does export assemble the BSP ahead of time?

No. QSS keeps the editable map, images and materials in memory, but `_export_object` assembles the BSP when you invoke export. It reads the source BSP, checks and encodes changed images, builds palette-indexed mip levels, rebuilds the relevant lumps and writes the output.

That is why the new Ironwail test action uses the normal exporter to make a temporary BSP. It avoids the need to maintain a separate preassembled copy, while still testing the actual export result.

Next useful test: install v0.18.4 and try Focus on a face whose normal points away from your current view, toggle the matrix in Edit Mode, paint 64 versus 65 faces, then use Test in Ironwail without making a normal export first.

‹/›

You said:

Color Palette: Can i have an quake palette color picker as alternative to the
gradient standard one in paint mode? Ideally switchable between the two in some
way.

Also - you couldnt know that - but in another session i was working with a
branch of you on a more sophisticated color converter to the quake palette. I

just put it into a "related tools" folder on the github branch. Its core is
python and you should use it for converting to quake palette.

so blender had a shortcut for fps all the time? :D nevermind then.

focus works great now. But the occluder thing does only work in paint mode.
Should work in edit mode too.

layermatrix loos great!

Selected all in a map and starte texture paint and got receted. I was more
looking for a kind of are you shure popup. Also is it possible to predict how
many would be possible before getting out of memory. 64 was a random value i
chose. There is much in the statiscs tab so maybe we can use this data to warn
more precise?

[Test in ironwail] works great. We dont need the other one anymore.

And you did not react to the image i sent concerning the flat color in edit
mode and its absence in object.

do we need a way to close maps?

Stopped thinking >

You said:

just uploaded the chatlog for the color converter so you understand what leda to it.

Worked for 3m 52s >

ChatGPT said:

Found it in the feature branch:

quake texture converter chat log.pdf ↗

I can see the file and its location, but I haven't successfully read the PDF contents yet. The GitHub file connector returned a binary-decoding error, and the PDF preview wouldn't load. I don't want to pretend I've understood the discussion without reading it.

Please attach the PDF directly here in the chat. Then I can go through the design history, understand why the converter uses its particular OKLab, segmentation and ramp-selection approach, and integrate the actual intended algorithm into QSS rather than approximating it.

You said:

PDF **quake texture co...** PDF

Worked for 3m 12s

ChatGPT said:

I've now read the chatlog. It fills in an important piece of the design history: **the converter wasn't intended to be a conventional nearest-color quantizer. Its priority became coherent, convincing Quake-style game textures, even when that means deliberately shifting some source colors.**

I understand the progression now.

What led to the current algorithm

1. **OKLab perceptual matching:** The initial nearest-palette approach produced noisy results. Floyd–Steinberg dithering made the noise worse, so dithering was deliberately dropped as the default.
2. **Adaptive hue mapping:** The next experiment allowed source colors to shift toward better-supported Quake colors. It was useful, but treating every color independently still produced speckling.
3. **Very light edge-aware smoothing:** Medium and strong smoothing lost too much surface detail. You preferred the light setting, around `0.025`.
4. **Palette ramps:** Colors were grouped into coherent shading families instead of being matched against an unordered collection of palette entries.

5. **Hierarchical material zones:** SLIC superpixels were merged into larger regions. Lighting differences were deliberately down-weighted so painted shadows and highlights wouldn't be mistaken for separate materials.
6. **Multiple candidate ramps per zone:** The first zone-based version became too brown because five ramps were excluded and each zone effectively had too little color diversity. The successful revision restored all 14 ramps and allowed up to three candidate ramps per material zone. quake texture converter chat log.pdf quake texture converter chat log.pdf quake texture converter chat log.pdf

The design priorities I'll preserve

- Avoid isolated color speckles and inconsistent shading within a surface.
- Preserve genuine material boundaries.
- Preserve painted highlights, shadows, scratches and fine detail.
- Allow some hue and saturation shifts in exchange for a convincing Quake look.
- Don't mistake low-resolution texture patterns or lighting variations for semantic material boundaries.
- Keep the source texture and diagnostic layers available for comparison.
- Keep transparency independent of color conversion, with index 255 reserved for transparent pixels.
- Don't assume that Soldier of Fortune's fullbright implementation is identical to Quake's. That investigation was explicitly left open. quake texture converter chat log.pdf

One unresolved detail: the chat ends with the observation that blue material areas can still be mapped too often to grey. Increasing `ramp_candidates_per_zone` from 3 to 5 was proposed as an experiment, but I don't see a confirmed result in the uploaded conversation. I won't treat that experiment as an established improvement.

I'll use the current `related tools/quake_texture_converter_fullbright_gui/conv_layers.py` as the implementation reference, and the chatlog as the source of the design rationale. I won't replace its algorithm with a simpler approximation or silently reintroduce the earlier experimental hue-shifting approach.

That gives me a much clearer target for integrating it into QSS.

You said:

yes 3 to 5 did the job! And glad youll integrate the converter. I planned to bring it into blender anyways to render stuff out with quake palette. Good job! Eager for the next version.

Worked for 14m 54s

ChatGPT said:

QSS v0.19.0 — CI passed

The next version is ready for Blender 5.2 testing.

Download Quake Surface Studio v0.19.0 ↗

SHA-256: `06704fd58a5110dc364474c5a0e981b4d40c12f093d7c9aa915bf43ef433172b`

What's new

Quake palette painting

Integrated

The new **QUAKE PALETTE** panel switches between Blender's Standard brush picker and the active map's actual 256-color Quake palette. Switching maps updates the palette source.

Your texture converter, inside Blender

Integrated

I brought over the existing material-zone converter rather than writing a substitute. It retains the OKLab, SLIC, hierarchical material merging, all 14 ramps, transparency and optional fullbright-mask workflow.

Most importantly, the default is now **five candidate ramps per material zone**, matching the setting you confirmed works.

The new **Convert to Quake Palette…** action can process:

- The active Texture Paint image.
- An image selected in the Image Editor.
- The current Render Result.

It writes a new indexed PNG and can optionally create the layered ORA diagnostic project. The source image is not overwritten.

One setup requirement: the converter needs NumPy, Pillow, SciPy and scikit-image. It runs through a separate Python environment, not an assumed Blender installation dependency.

On Linux, create it with:

```
python3 -m venv ~/.local/share/qss/converter-venv
~/.local/share/qss/converter-venv/bin/python -m pip install --upgrade pip
~/.local/share/qss/converter-venv/bin/python -m pip install numpy Pillow scipy
scikit-image
```

Then enter this executable in **QSS Settings → Quake converter Python**:

```
~/.local/share/qss/converter-venv/bin/python
```

Use **Check Converter Python** in the new panel to verify it.

Painting and navigation

The old hard 64-face rejection is gone. Large selections now get a confirmation dialog showing estimated image-buffer allocation, existing map image memory and available system memory where detectable. It is an estimate, not a guarantee against Blender running out of memory.

The foreground clipping change now applies in **Edit Mode as well as Texture Paint**.

Map management and viewport shading

There is now a **Close** button beside Loaded Map. It confirms before removing the map and its orphaned map-only materials/images, preserves shared assets, and restores non-QSS scene objects when the final map is closed.

I also addressed the screenshot. It looks likely that the two viewports are using different shading modes: the left displays material preview, while the right appears to use Solid shading. These are viewport settings, not separate Object Mode and Edit Mode color spaces.

QSS now shows a hint and a **Material Preview** shortcut when the active viewport is in Solid mode.

The separate Last Export button has also been removed. **Test in Ironwail** is now the single test-launch workflow.

Validation

176 tests passed

The CI run installed the converter dependencies, compiled the Python modules, ran the standalone tests—including an actual conversion to an indexed PNG—validated the version, assembled the Blender package and uploaded the artifact.

Interactive Blender 5.2 testing is still needed for the palette UI, render conversion, focus clipping and map-close behavior.

Draft PR #1 — QSS v0.19.0 ↗ is updated and remains unmerged. `main` is untouched.

I'm particularly interested in how the five-ramp converter feels when used directly on your Blender paint images and renders. This is the first step toward making QSS a useful Quake texture authoring tool, rather than only a BSP editor.

You said:

aput a FOV SETTING in the config please.

The quake palette does not open and in the right click menu in paint mode we
get just the original color picker.

About the texture conversion: I dont want that as another section. I also dont
want another external tool tbh. If python them i want it to work inside
blender. But.. ya its blender: I'd bet you would be able to implement it as a
shader.

tryed the warning system. It warned me i gnored, it crashed and threw me back

to edit mode. Good job. Maybe a little bit more understandable for everyone the
warning. :D

You are wrong about the viewport shading. Both sreenshots are taken in
"material preview" what you thought would be the solid workbank is eevee with
thre weird greyish tint that only is drawn on edit mode. It also is there in
cycles. And it is not there before i import a map. Even when i copy the whole
map and paste it in a new blender file edit mode looks normal

I'll treat these as fixes to the v0.19.0 branch, not as a new UI expansion. I'm checking Blender 5.2's actual paint-palette API and the imported material/shader setup first; the grey Edit Mode tint needs a root-cause fix, not another guess about viewport shading. I'll also move conversion into Blender's Python execution path and make the large-selection warning explain what it is estimating and what can happen if you continue.

Inspected implementation details

Explored palette and field of view

Searched the web

Planned implementation and refined the design

Integrating palette access

ChatGPT said:

QSS v0.20.0 — CI passed

Download QSS v0.20.0 for Blender 5.2 ↗

SHA-256: `520ef77fda16ef3e7020c963a3de250de731e5b37dab104ef9c027f56ca3e199`

What changed

1. Configurable FOV

Added **Viewport FOV** to QSS Settings. It defaults to **90° horizontal FOV**, persists in `config.json`, applies to open 3D Views, and is also used by Focus on Face.

2. Quake palette picker fixed

I found two problems: the palette creation path was missing its `hashlib` import, and the palette UI used an unsupported argument and omitted Blender's palette selector.

The picker now uses Blender's native palette selector and swatch grid. The Standard / Quake switch is inside the existing **LIGHTING** panel, and there's a **Quake Palette Colors...** entry in the paint-mode right-click menus. The regular Blender color picker remains available.

raw.githubu... +1

3. Conversion now runs inside Blender

No external executable, no separate Python environment and no NumPy/Pillow/SciPy installation.

The converter now runs from `palette_utils.py` inside Blender's Python process. It retains the material-zone approach, all 14 ramps and your confirmed **five candidate ramps per zone**. It uses the active map's actual palette and creates a new indexed PNG without overwriting the source.

I didn't implement this as a shader: a shader can map individual pixels to palette colors, but it can't reproduce the converter's whole-image superpixel segmentation, connected material-zone merging and per-zone ramp fitting. The Python pass is the appropriate place for that work.

One caveat: the pure-Python implementation has not yet been benchmarked on large renders in Blender. It may be noticeably slower than the previous NumPy/scikit-image version.

4. Paint warning improved

The arbitrary 64-face threshold is gone. QSS estimates the face images and atlas buffers from the selected faces' UV footprints, adds a conservative working-memory reserve, and compares that estimate with available memory.

The warning now explains the crash risk in plain language and recommends smaller batches. It still cannot guarantee that Blender won't run out of memory if you choose to continue.

5. Edit Mode grey tint

You're right: my previous Solid-shading explanation was wrong.

I now target Blender's **face-fill selection overlay** rather than changing the viewport shading mode. QSS temporarily disables that overlay while a QSS map is in Edit Mode and restores the previous setting outside Edit Mode. Blender exposes this as `View3DOverlay.show_faces`. Blender Python...

This is a targeted fix for the behavior you described, but I need your local Blender test to confirm that it removes the grey tint.

Validation and next test

181 tests passed

CI verified Python compilation, the in-process converter, palette transparency/fullbright handling, indexed PNG output, FOV calculations and package assembly. It does not run Blender's interactive GUI.

The updated draft PR #1 ↗ remains unmerged. `main` is untouched.

The most useful local test now is: enter Texture Paint, right-click the brush color and choose **Quake Palette Colors...**, convert a small texture, then import a map and check whether the Edit Mode tint has disappeared.

You've reached the maximum length for this conversation, but you can keep talking by starting a new chat.